STT 843: Multivariate Analysis

7. Principal Component Analysis (Chapter 8.1)

Guanqun Cao

Department of Statistics and Probability
Michigan State University

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Outline

1. Population Principle Components
2. Singular Value Decomposition (SVD)

What way should you tilt your shark head?

- Imagine you are a whale shark with a wide mouth and that you're swimming towards a delicious swarm of krill.
- What way should you tilt your shark head in order to eat as many krill as possible? The answer is given by the first principal component of the data!

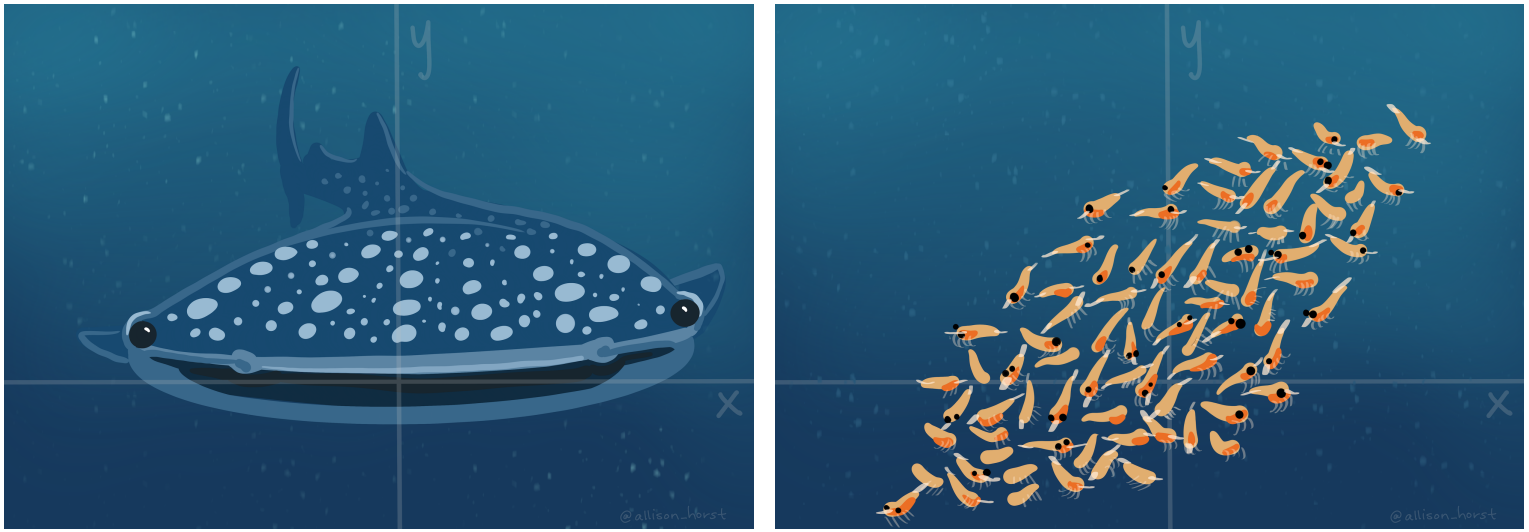

Figure: Credit to Allison Horst (allison-horst)

Informal introduction

The name principal component analysis comes from thinking of this as splitting the data $\boldsymbol{X}$ into its most important parts. It therefore won't surprise you to find that this involves the matrix decompositions.

With multivariate data, it is common to want to reduce the dimension of the data in a sensible way. For example

- exam marks across different modules are averaged to produce a single overall mark for each student
- a football league table converts the numbers of wins, draws and losses to a single measure of points.

Mathematically, these summaries are both linear combinations of the original variables of the form

$$Y = \boldsymbol{u}^T X$$

for some choice of u.

A single $\boldsymbol{u}$ gives a snapshot or summary of the data. If $\boldsymbol{u}$ is chosen well that snapshot may tell us much of what we want to know about the data, e.g.,

- Liverpool won the league,
- student X's exam performance was first class etc.

What is PCA?

A principle component analysis is concerned with explaining the variance-covariance structure of a set of variables through a few linear combinations of these variables. It general objectives are

1. data reduction

   Although p components are required to reproduce the total system variability, often much of this variability can be accounted for by a small number k of the principle components.

2. interpretation

   An analysis of principle components often reveals relationships that were not previously suspected and thereby allows interpretations that would not ordinarily results.

- Seek to maximize the variance of a linear combination of the variables
- Purpose: explain the covariance structure of a set of variables (one-sample technique)
- Often useful as inputs to another analysis
- **Principal Component Regression** involves converting p collinear predictors into $k < p$ independent components
- Often yields better estimates of regression coefficients

Population Principle Components

- Algebraically, principal components are particular linear combinations of the p random variables $X_1, X_2, \ldots, X_p$.
- Geometrically, these linear combination represent the selection of a new coordinate system obtained by rotating the original system with $X_1, X_2, \ldots, X_p$ as the coordinate axes.
- The new axes represents the directions with maximum variability and provide a simpler and more parsimonious description of the covariance structure.

Population Principle Components and Covariance matrix

- Principal components depend solely on the covariance matrix $\boldsymbol{\Sigma}$ (or the correlation matrix $\boldsymbol{\rho}$) of $X_1, X_2, \ldots, X_p$. Their development does not require a multivariate normal assumption.
- On the other hand, principal components derived for multivariate normal populations have useful interpretations in terms of the constant density ellipsoids.

Notation

As before, let $\mathbf{x}_1, \ldots, \mathbf{x}_n$ be $p \times 1$ vectors of measurements on n experimental units and write

$$\mathbf{X} = \begin{pmatrix} -\mathbf{x}_1^\top - \\ -\mathbf{x}_2^\top - \\ \cdots \\ -\mathbf{x}_n^\top - \end{pmatrix} \tag{1}$$

NOTE:

- In this section we will assume that $\mathbf{X}$ has been column centered so that the mean of each column is 0 (i.e., the sample mean of $\mathbf{x}_1, \ldots, \mathbf{x}_n$ is the zero vector $\mathbf{0} \in \mathbb{R}^p$).
- If $\mathbf{X}$ has not been column centered, replace $\mathbf{X}$ by $\mathbf{HX}$ where $\mathbf{H}$ is the centering matrix $(\boldsymbol{H} = \boldsymbol{I}_n - n^{-1}\mathbf{1}_n\mathbf{1}_n)$, or equivalently, replace $\mathbf{x}_i$ by $\mathbf{x}_i - \bar{\mathbf{x}}$.
- It is possible to write out the details of PCA replacing $\mathbf{X}$ by $\mathbf{HX}$ throughout, but this gets messy and obscures the important detail.
- Most software implementations (and in particular `prcomp` in R), automatically center your data for you.

The sample covariance matrix for $\mathbf{X}$ (assuming it has been column centered) is

$$\mathbf{S} = \frac{1}{n}\mathbf{X}^\top\mathbf{X} = \frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{x}_i^\top \tag{2}$$

Given some vector $\mathbf{u}$, the transformed variables $y_i = \mathbf{u}^\top\mathbf{x}_i$ have mean 0:

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i = \frac{1}{n}\sum_{i=1}^{n}\mathbf{u}^\top\mathbf{x}_i = \frac{1}{n}\mathbf{u}^\top\sum_{i=1}^{n}\mathbf{x}_i = 0 \tag{3}$$

as the mean of the $\mathbf{x}_i$ is 0.

The sample covariance matrix $\mathbf{u}^\top \mathbf{S}\mathbf{u}$ is:

$$\frac{1}{n}\sum_{i=1}^{n} y_i^2 = \frac{1}{n}\sum_{i=1}^{n} \mathbf{u}^\top \mathbf{x}_i \mathbf{x}_i^\top \mathbf{u} = \frac{1}{n}\mathbf{u}^\top \sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i^\top \mathbf{u} = \mathbf{u}^\top \mathbf{S}\mathbf{u} \tag{4}$$

- The spectral decomposition theorem gives a decomposition of any symmetric matrix. We now give a generalisation of this result which applies to all matrices.
- If matrix $\mathbf{A}$ is not a square matrix, then it cannot have eigenvectors. Instead, it has singular vectors corresponding to singular values. Suppose $\mathbf{A}$ is an $n \times p$ matrix. Then we say σ is a singular value with corresponding left and right singular vectors $\mathbf{u}$ and $\mathbf{v}$ (respectively) if

$$\mathbf{A}\mathbf{v} = \sigma\mathbf{u} \quad \text{and} \quad \mathbf{A}^\top\mathbf{u} = \sigma\mathbf{v} \tag{5}$$

- If $\mathbf{A}$ is a symmetric matrix then $\mathbf{u} = \mathbf{v}$ is an eigenvector and σ is an eigenvalue.

Singular value decomposition

Let $\mathbf{A}$ be an $n \times p$ matrix of rank r, where $1 \leq r \leq \min(n, p)$. Then there exists an $n \times r$ matrix $\mathbf{U} = [\mathbf{u}_1, \ldots, \mathbf{u}_r]$, a $p \times r$ matrix $\mathbf{V} = [\mathbf{v}_1, \ldots, \mathbf{v}_r]$, and an $r \times r$ diagonal matrix $\mathbf{\Sigma} = \text{diag}\{\sigma_1, \ldots, \sigma_r\}$ such that

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top \tag{6}$$

where $\mathbf{U}^\top\mathbf{U} = \mathbf{I}_r = \mathbf{V}^\top\mathbf{V}$ and $\sigma_1 \geq \cdots \geq \sigma_r > 0$.

Non-compact form of SVD

Sometimes we write it in a non-compact form $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\top$ where $\mathbf{U}$ is an $n \times n$ orthogonal matrix ($\mathbf{U}^\top\mathbf{U} = \mathbf{U}\mathbf{U}^\top = \mathbf{I}_n$), $\mathbf{V}$ is a $p \times p$ orthogonal matrix ($\mathbf{V}^\top\mathbf{V} = \mathbf{V}\mathbf{V}^\top = \mathbf{I}_p$), and Σ is an $n \times p$ diagonal matrix:

$$\Sigma = \begin{pmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & 0 & \dots \\ \vdots & 0 & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_r \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix} \tag{7}$$

Let $\mathbf{A}$ be any matrix of rank r. Then the non-zero eigenvalues of both $\mathbf{A}\mathbf{A}^\top$ and $\mathbf{A}^\top\mathbf{A}$ are $\sigma_1^2, \ldots, \sigma_r^2$. The corresponding unit eigenvectors of $\mathbf{A}\mathbf{A}^\top$ are given by the columns of $\mathbf{U}$, and the corresponding unit eigenvectors of $\mathbf{A}^\top\mathbf{A}$ are given by the columns of $\mathbf{V}$.

Example

Let $\mathbf{A} = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix}$. The singular values are the square roots of the eigenvalues of $\mathbf{A}\mathbf{A}^\top$:

$$\mathbf{A}\mathbf{A}^\top = \begin{pmatrix} 17 & 8 \\ 8 & 17 \end{pmatrix} \tag{8}$$

Solving $\det(\mathbf{A}\mathbf{A}^\top - \lambda \mathbf{I}) = (17-\lambda)^2 - 64 = 0$ gives $\lambda_1 = 25$ and $\lambda_2 = 9$. Thus the singular values are $\sigma_1 = 5$ and $\sigma_2 = 3$.
The compact SVD is:

$$\mathbf{A} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 5 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{3\sqrt{2}} & -\frac{1}{3\sqrt{2}} & \frac{4}{3\sqrt{2}} \end{pmatrix} \tag{9}$$

Geometric Interpretation

We think of PCA as projecting the data points $\mathbf{x}$ onto a subspace U. The basis vectors for this subspace are the eigenvectors of $\boldsymbol{S}$, which are the same as the right singular vectors of $\mathbf{X}$ (the loadings):

$$U = \text{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_r\} \tag{10}$$

The orthogonal projection matrix for projecting onto U is:

$$P_U = \mathbf{U}\mathbf{U}^\top \tag{11}$$

since $\mathbf{U}^\top\mathbf{U} = \mathbf{I}$.

The coordinates of the data points projected onto V (with respect to the basis for U) are the principal component scores:

$$\mathbf{y}_i = \begin{pmatrix} y_{i1} \\ \vdots \\ y_{ir} \end{pmatrix} = \mathbf{U}^\top \mathbf{x}_i \tag{12}$$

where $\mathbf{U} = (\mathbf{u}_1 \dots \mathbf{u}_r)$ is the matrix of right singular vectors from the Singular Value Decomposition (SVD) of $\mathbf{X}$.

The transformed variables (matrix of scores) are:

$$\mathbf{Y} = \begin{pmatrix} -\mathbf{y}_1^\top - \\ \vdots \\ -\mathbf{y}_n^\top - \end{pmatrix} = \mathbf{X}\mathbf{U} \tag{13}$$

Substituting the SVD for $\mathbf{X} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$, we can see the transformed variable matrix/principal component scores are:

$$\mathbf{Y} = \mathbf{U}\boldsymbol{\Sigma} \tag{14}$$

$\mathbf{Y}$ is an $n \times r$ matrix, and so if $r < p$ we have reduced the dimension of $\mathbf{X}$, keeping the most important parts of the data.

Let the random vector $\boldsymbol{X}^{\mathsf{T}} = [X_1, X_2, \ldots, X_p]$ have the covariance matrix $\boldsymbol{\Sigma}$ with eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p \geq 0$.
Consider the linear combinations

$$
\begin{aligned}
Y_1 &= \mathbf{a}_1^{\mathsf{T}} \boldsymbol{X} = a_{11} X_1 + a_{12} X_2 + \cdots + a_{1p} X_p \\
Y_2 &= \mathbf{a}_2^{\mathsf{T}} \boldsymbol{X} = a_{21} X_1 + a_{22} X_2 + \cdots + a_{2p} X_p \\
&\vdots \\
Y_p &= \mathbf{a}_p^{\mathsf{T}} \boldsymbol{X} = a_{p1} X_1 + a_{p2} X_2 + \cdots + a_{pp} X_p
\end{aligned}
$$

Then

$$
\begin{aligned}
\operatorname{Var}(Y_i) &= \mathbf{a}_i^{\mathsf{T}} \boldsymbol{\Sigma} \mathbf{a}_i \quad i = 1, 2, \ldots, p \\
\operatorname{Cov}(Y_i, Y_k) &= \mathbf{a}_i^{\mathsf{T}} \boldsymbol{\Sigma} \mathbf{a}_k \quad i, k = 1, 2, \ldots, p
\end{aligned}
$$

Define

- First principle component = linear combination $\mathbf{a}_1^\top \boldsymbol{X}$ that maximizes $\mathrm{Var}\left(\mathbf{a}_1^\top \boldsymbol{X}\right)$ subject to $\mathbf{a}_1^\top \mathbf{a}_1 = 1$
- Second principle component = linear combination $\mathbf{a}_2^\top \boldsymbol{X}$ that maximizes

$$\mathrm{Var}\left(\mathbf{a}_2^\top \boldsymbol{X}\right) \text{ subject to } \mathbf{a}_2^\top \mathbf{a}_2 = 1 \text{ and}$$
$$\mathrm{Cov}\left(\mathbf{a}_1^\top \boldsymbol{X}, \mathbf{a}_2^\top \boldsymbol{X}\right) = 0$$

- At the i th step,
  i th principle component = linear combination $\mathbf{a}_i^\top \boldsymbol{X}$ that maximizes

$$\mathrm{Var}\left(\mathbf{a}_i^\top \boldsymbol{X}\right) \text{ subject to } \mathbf{a}_i^\top \mathbf{a}_i = 1 \text{ and}$$
$$\mathrm{Cov}\left(\mathbf{a}_i^\top \boldsymbol{X}, \mathbf{a}_k^\top \boldsymbol{X}\right) = 0 \text{ for } k < i$$

Assuming $\mathbf{x}_i, i = 1, \ldots, n$, are centered at $\mathbf{0}$, we rotate the axes by multiplying $\mathbf{x}_i$ by an orthogonal matrix $\mathbf{A}$:

$$\mathbf{y}_i = \mathbf{A}\mathbf{x}_i$$

- Distance from origin is same for both old and new data

$$\mathbf{y}_i^\top \mathbf{y}_i = (\mathbf{A}\mathbf{x}_i)^\top (\mathbf{A}\mathbf{x}_i) = \mathbf{x}_i^\top \underbrace{\mathbf{A}^\top \mathbf{A}}_{\substack{\mathbf{A} \text{ is} \\ \text{orthogonal}}} \mathbf{x}_i = \mathbf{x}_i^\top \mathbf{x}_i$$

- If $\mathbf{A}$ were chosen to rotate axes in harmony with ellipsoid, then

$$\text{var}\{\mathbf{y}\} = \mathbf{S}_y = \mathbf{A}\mathbf{S}\mathbf{A}^\top = \begin{bmatrix} s_{y_1}^2 & & & \mathbf{0} \\ & s_{y_2}^2 & & \\ & & \ddots & \\ \mathbf{0} & & & s_{y_p}^2 \end{bmatrix}$$

- But, by definition, the matrix $\mathbf{A}$ which diagonalizes $\mathbf{S}$ is a matrix containing the normalized eigenvectors of $\mathbf{S}$.
- $y_{i1} = \mathbf{a}_1^{\mathsf{T}}\mathbf{x}_i, y_{i2} = \mathbf{a}_2^{\mathsf{T}}\mathbf{x}_i$, etc. are the "principal component scores" for subject i, where $\text{var}\{y_{ij}\} = s_{y_j}^2 = \lambda_j$ (the jth ordered eigenvalue of $\mathbf{S}$).
- Since $\mathbf{a}_1$ specifies the single dimension of greatest separation among the data, it is expected that $\text{var}\{y_1\}$ should have largest variance among $\left(s_{y_j}^2\right)$ and $\text{var}\{y_p\}$ should have smallest variance.
- Number of "significant" eigenvalues indicates number of "essential dimensions" in data.

Principal components are NOT scale invariant

- Orientation of P.C.'s changes if we convert from: inches to centimeters, pounds to kg, original scale to standardized scale, etc. ex

- $$\mathbf{S} = \begin{bmatrix} 1 & 2 \\ 2 & 100 \end{bmatrix} \quad \begin{array}{c}\text{convert}\\ \text{to certain}\\ \text{matrix}\end{array} \quad \mathbf{R} = \begin{bmatrix} 1 & .2 \\ .2 & 1 \end{bmatrix}$$
- $\lambda_1 = 100.04, \quad \lambda_2 = 0.96, a_1 = [.0202, .9998], a_2 = [.9998, -.0202], \frac{\lambda_1}{\lambda_1+\lambda_2} = 99.04\%,$
- $\lambda_1^{(R)} = 1.2, \lambda_2^{(R)} = 0.8,$
- $a_1^{(R)} = [.7071, .7071], a_2^{(R)} = [.7071, -.7071]$
- $\frac{\lambda_1^{(R)}}{\lambda_1^{(R)}+\lambda_2^{(R)}} = 60\%$

No simple relationships exist between:

- λ_i and $\lambda_i^{(R)}$
- $\frac{\lambda_i}{\Sigma\lambda_i}$ and $\frac{\lambda_i^{(R)}}{\Sigma\lambda_i^{(R)}}$
- $\mathbf{a}_1$ and $\mathbf{a}_1^{(R)}$